DO NOW — RECALL

Programming Techniques · Iteration and Tracing

Name: _____ Class: _____ Date: _____

1. What prints after int x=2; x=x*3;?

2. What does a queue remove first?

3. Why should loops have a stopping condition?

4. What is 0 XOR 1?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

6

2. Model answer

The oldest item

3. Model answer

To avoid non-termination

4. Model answer

1

By the end, you can…

• Distinguish count-controlled and condition-controlled loops

• Trace variables across loop iterations

• Write a terminating C# loop

Vocabulary — know these terms

Recall the term — iteration

Recall the term — accumulator

Recall the term — counter

Recall the term — condition

Recall the term — termination

NEW LEARNING

Read, annotate and explain

Trace an accumulator

int total = 0; for (int i = 1; i <= 3; i++) total += i;

Explain the important decision

Initialise the accumulator once. Each iteration changes total; i++ then advances the counter. Test i <= 3 before every pass.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Trace an accumulator

Initialise and trace pass one

Initially total=0 and i=1. The test 1<=3 is true; total becomes 1, then i becomes 2.

Trace the remaining passes

At i=2, total becomes 1+2=3; i becomes 3. At i=3, total becomes 3+3=6; i becomes 4.

Show the failed test

4<=3 is false, so the body stops and total remains 6. A while loop may run zero times; a do...while tests after its body and runs at least once.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

1. Problem 1

for i = 0 to 3 exclusive, how many iterations?

Method / working:

2. Problem 2

while (x < 5) x++; starting x=3: final x?

Method / working:

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Apply independently
Trace the final total for 2 + 4 + 6 using an accumulator. [3]

Question 2

Apply independently
Give a reason to choose a while loop rather than for. [2]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Apply independently
Write a C# loop to output 1, 2, 3. [3]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

True or false: a while loop always runs at least once.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.